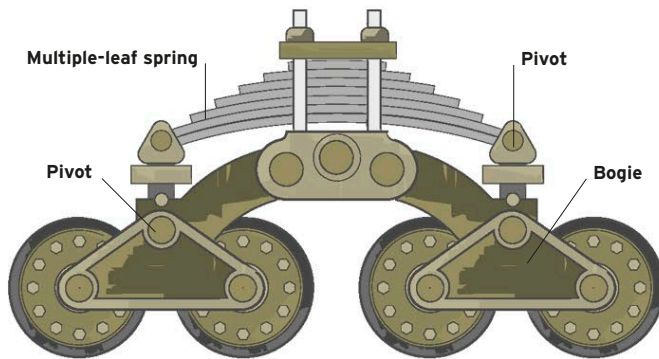


TYPES OF TANK SUSPENSION

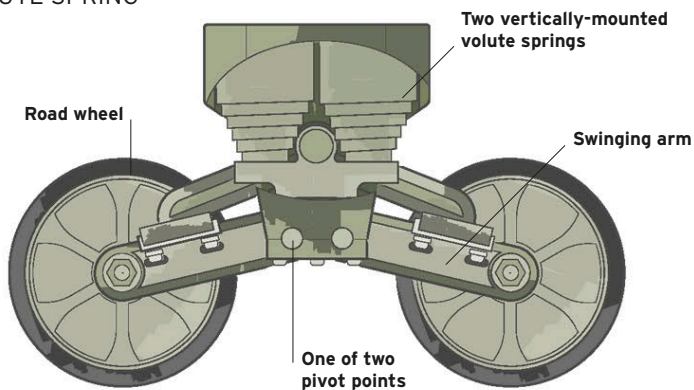
There have been six successful suspension systems employed in tracked armored fighting vehicles, and several more that were abandoned. Of the successful ones, five relied on the most significant physical property of spring steel: its determination to return to the form in which it was manufactured

LEAF SPRING



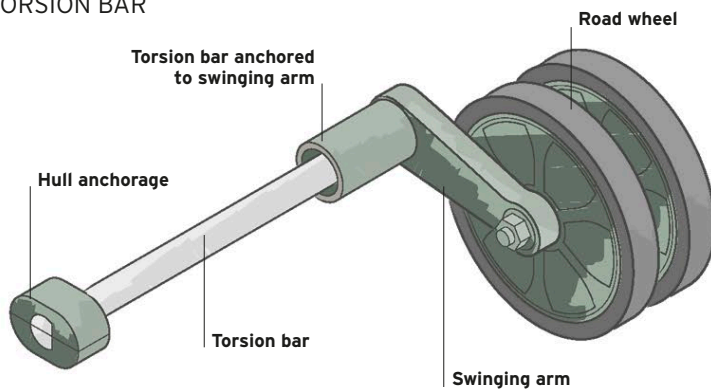
In use since medieval times, leaf springs are the simplest form of sprung suspension. Strips, or “leaves,” of arced, highly resistant steel are stacked together and mounted so that they absorb the upward pressure of a wheel, pair of wheels, or pair of wheel bogies (as above), and then return to their original configuration.

VOLUTE SPRING



A volute spring is a coil-wound leaf spring, the center of which is then pulled out to form a truncated cone. It acts in compression, the coils sliding over each other, and can be mounted vertically (as above) or horizontally. Volute springs were commonly mounted in tandem pairs on a bogie; road wheels acted on the springs by way of swinging arms.

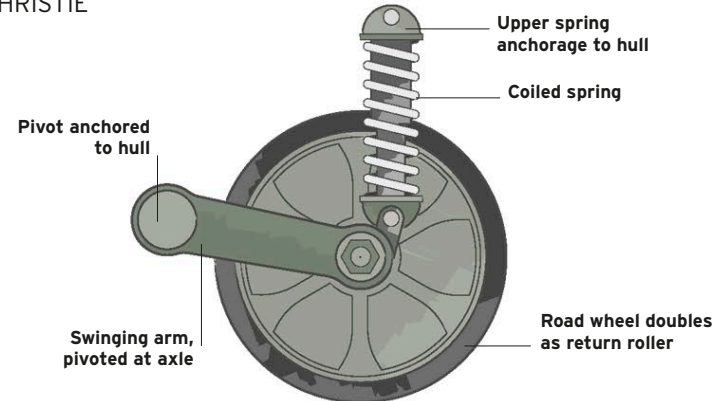
TORSION BAR



Torsion bar suspension also relies on the “memory” of yield-resistant spring steel to maintain its original configuration—in this case as a rod anchored at one end to the tank’s chassis. As its name suggests, the pressure takes the form of a twisting motion imparted by an arm connecting the rod’s free end to the road wheel’s axle.

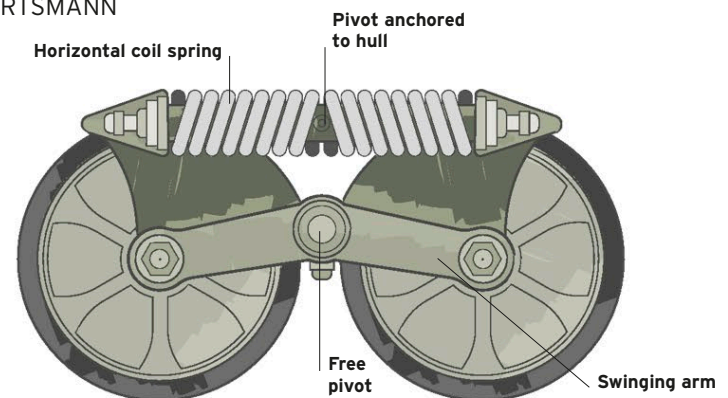
at the earliest possible opportunity. The most effective of these “spring” systems is the torsion bar, which is the only one still in widespread use today. The sixth system is the active hydropneumatic arrangement, which was first employed in Citroën passenger cars in the mid-1950s.

CHRISTIE



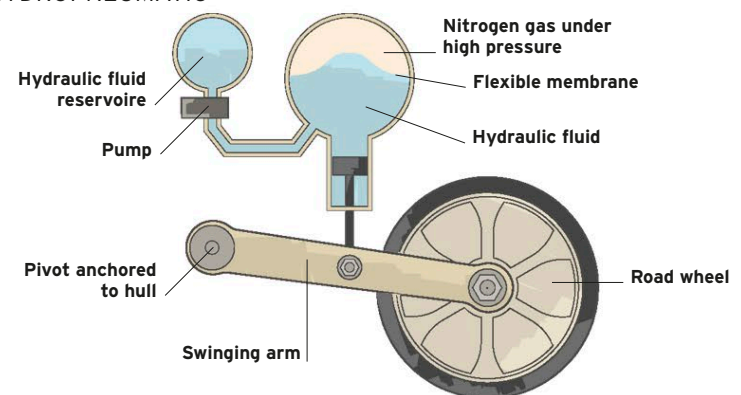
Developed by J. Walter Christie as part of his efforts to improve overall tank design, this simple system incorporated a coiled spring, which he first mounted vertically, although later horizontal versions proved to be more effective. The large-diameter road wheels acted as return rollers and were mounted in pairs with the track guides running between them.

HORTSMANN



In the Horstmann system, paired road wheels are mounted on swinging arms, the upward motion of which is cushioned by the compression of springs mounted horizontally between them. It is similar to the horizontal volute-spring system, but improves on it: unlike the volute spring, the coiled spring operates in both extension and compression, and so increases wheel travel.

HYDROPNEUMATIC



In this system, each road wheel is attached to a sphere containing two chambers—an upper one containing nitrogen gas under high pressure, and a lower one containing hydraulic fluid—with a flexible membrane in between. A pump pressurizes the fluid, to which additional pressure is applied from the road wheel when under load; the gas compresses, thus acting as a spring.